

DSC 335 Exam 2 QMD Part

Name:

Question 1 (13 points)

The *births14* data set in the **openintro** R package gives information on a random sample of human births recorded by the US Department of Health and Human Services in 2014. Our response variable is the *weight* of the babies, in pounds. For this problem we will consider the following predictors: the *sex* of the baby (male or female, with female as the baseline level), the mother's age (*mage*), and the number of *weeks* the mother was pregnant before giving birth.

a. (6) Consider the following prior information: For children of the same sex and mother's age, we think that for each additional week of pregnancy the weight of the baby will increase by 0.5 pounds, but that the increase could plausibly be anywhere from 0.1 to 0.9 pounds. We also think that male babies will weigh around 0.4 pounds more than female babies born at the same number of weeks to mothers of the same age, and that this difference could plausibly be between 0.2 and 0.6 pounds. Finally, we think that a $N(0, 1)$ distribution is a reasonable prior for the coefficient of the *mage* variable. Using this prior information and putting default priors on all other parameters, fit a Normal Bayesian regression model in the code chunk below. (If you are unable to figure out how to incorporate the priors, you can get partial credit for part (a) but potentially full credit for the rest of the question if you run the code with default priors.)

```
exam_mod1 <- stan_glm(weight~sex+mage+weeks, data=births14,
                      family=gaussian,
                      prior=normal(location=c(0.4,0,0.5),
                                           scale=c(0.1,1,0.2)),
                      chains=4, iter=5000*2,seed=84735)
```

b. (4) Based on 90% posterior credible intervals, which of the variables are significant predictors of a baby's weight, given the other variables in the model. Justify your answer.

```
tidy(exam_mod1, effects=c("fixed", "aux"),
     conf.int = TRUE, conf.level=.9)
```

```
# A tibble: 6 x 5
  term          estimate std.error conf.low conf.high
  <chr>          <dbl>     <dbl>   <dbl>   <dbl>
1 (Intercept)  -4.37      0.542   -5.27    -3.47
2 sexmale       0.390     0.0556   0.299    0.482
3 mage         0.0169    0.00582  0.00729   0.0265
4 weeks        0.282     0.0131   0.260    0.303
5 sigma        1.07      0.0243   1.04     1.12
6 mean_PPD     7.20      0.0481   7.12     7.28
```

All three variables are significant predictors of a baby's weight, given the other variables in the model, because the posterior credible intervals are all above 0.

c. (3) Calculate a 95% posterior predictive interval for the weight of a male baby born to a 30 year old mother whose pregnancy lasted 37 weeks.

```
postpred <- posterior_predict(exam_mod1,
                             newdata=data.frame(sex="male",mage=30,weeks=37))
posterior_interval(postpred, prob=.95)
```

```
      2.5%    97.5%
1 4.863495 9.05648
```

Question 2 (10 points)

The *q3data* data set created in the code chunk below contains the variable *median_edu* which gives the median education level of counties in the US. It has four levels: *below_hs*, *hs_diploma*, *some_college*, and *bachelors*. It also contains the following predictors: the percent of the population below the *poverty* line, the population of the county in 2017 (*pop2017*), and *metro*, which is yes if the county has a metro area and no if it does not. We will use the *q3data* to classify the median education level of Boyle County, which had a 2017 population of 29924, a poverty level of 15.5, and does not have a metro area.

```
q2data <- usdata::county %>% select(median_edu,metro,poverty,pop2017) %>% na.omit()
```

a. (3) Fit a Naive Bayes model that only includes the *poverty* and *pop2017* variables. How would this model classify the median education level of Boyle County?

```
edu_mod1 <- naiveBayes(median_edu~poverty+pop2017, data=q2data)
ourdata=data.frame(metro="no", poverty=15.5, pop2017=29924)
predict(edu_mod1, newdata=ourdata)
```

```
[1] hs_diploma
Levels: below_hs hs_diploma some_college bachelors
```

We would classify Boyle County as having a median education level of `hs_diploma`.

b. (3) Fit a Naive Bayes model that includes the *poverty*, *pop2017*, and *metro* variables. How would this model classify the median education level of Boyle County?

```
edu_mod2 <- naiveBayes(median_edu~poverty+pop2017+metro, data=q2data)
predict(edu_mod2, newdata=ourdata)
```

```
[1] hs_diploma
Levels: below_hs hs_diploma some_college bachelors
```

We would classify Boyle County as having a median education level of `hs_diploma`.

c. (4) Now consider the in-sample (not cross-validated) performance of the two models above. Which did a better job of classifying the counties whose median education level is a `some_college`? Justify your answer.

```
naive_classification_summary(edu_mod1, data=q2data,y="median_edu")
```

```
$confusion_matrix
  median_edu  below_hs      hs_diploma some_college bachelors
  below_hs  0.00% (0) 100.00%      (2)  0.00% (0) 0.00      (0)
  hs_diploma 0.21% (3)  98.00% (1,369)  1.79% (25) 0.00
  some_college 0.65% (11) 76.58% (1,295) 22.77% (385) 0.00
  bachelors  2.17% (1)  36.96%      (17) 60.87% (28) 0.00

$overall_accuracy
[1] 0.5593112
```

```
naive_classification_summary(edu_mod2, data=q2data,y="median_edu")
```

```

$confusion_matrix
  median_edu  below_hs      hs_diploma some_college bachelors
below_hs 0.00% (0) 100.00%      (2) 0.00%      (0) 0.00      (0)
hs_diploma 0.36% (5) 94.42% (1,319) 5.23%      (73) 0.00      (3)
some_college 0.47% (8) 61.38% (1,038) 37.97% (642) 0.18      (2)
bachelors 0.00% (0) 26.09%      (12) 69.57%      (32) 4.35      (2)

$overall_accuracy
[1] 0.6259566

```

The second model correctly classified about 38% of counties with some_college as the median education level, considerably higher than the 23% from the first model.

Question 3 (17 points)

Dr. Daryl is concerned that students aren't attending his office hours enough so one semester he decides to track the number of students that come to his office during each 1 hour period. Prior to the semester starting, he thinks that the average number of students that come each hour could be modeled by a $\text{Gamma}(1, 0.5)$ distribution. He records the first 10 office hour attendances as follows: 1,3,0,4,0,2,2,0,3,1.

a. (5) In the code chunk below, fill in the missing information (wherever there is a ?) to perform an MCMC simulation of his posterior understanding of the true average number of students λ that show up to a single office hour, based on a Gamma-Poisson model.

```

# STEP 1: DEFINE the model
gp_model <- "
  data {
    int<lower = 0> Y[10];
  }
  parameters {
    real<lower = 0> lambda;
  }
  model {
    Y ~ poisson(lambda);
    lambda ~ gamma(1,0.5);
  }
"

# STEP 2: SIMULATE the posterior
gp_sim <- stan(model_code = gp_model, data = list(Y = c(1,3,0,4,0,2,2,0,3,1)),
              chains = 4, iter = 5000*2, seed = 84735)

```

Trying to compile a simple C file

b. (3) Dr. Daryl wants to use his posterior distribution to test the hypotheses $H_0 : \lambda \geq 2$ versus $H_a : \lambda < 2$. Calculate the posterior probability of both hypotheses.

```
gp_df <- as.data.frame(gp_sim, pars = "lp__", include = FALSE)
gp_df %>%
  mutate(postprob = lambda < 2) %>%
  tabyl(postprob)
```

postprob	n	percent
FALSE	3351	0.16755
TRUE	16649	0.83245

The posterior probability of H_0 is .16755 and of H_a is .83245.

c. (3) The code chunk below gives the prior probabilities of the hypotheses. Use these probabilities and your answers from part b to calculate the Bayes Factor for the alternative hypothesis.

```
pgamma(2,1,0.5)
```

```
[1] 0.6321206
```

```
1-pgamma(2,1,0.5)
```

```
[1] 0.3678794
```

```
(.83245/.16755)/(pgamma(2,1,0.5)/(1-pgamma(2,1,0.5)))
```

```
[1] 2.891474
```

d. (3) Summarize the strength of the evidence for the alternative hypothesis based on your answers to parts b and c.

The posterior probability of the alternative hypothesis is quite a bit higher than that of the null hypothesis but it is still only around 83% so that is not overwhelming evidence. The Bayes Factor is relatively small at 2.89 showing that our belief in the hypotheses have only slightly

changed from the prior beliefs. Putting the two together there is moderate evidence that the alternative is true, but we may want to gather more data to have a more concrete answer.

e. (3) Suppose our hypotheses had been $H_0 : \lambda = 2$ versus $H_a : \lambda \neq 2$ instead. Describe a method for conducting such a test (but do not actually perform the test).

We could have calculated a posterior credible interval for λ . If 2 is not contained in our interval, we would have strong evidence for the alternative. If not, the evidence for the alternative would be unconvincing. Alternatively, we could define a null hypothesis such as $H_0 : \lambda \in (2-k, 2+k)$ for some k chosen by content knowledge experts, and conducted a test similar to that above.

f. (Bonus) The MCMC procedure above produces 20,000 λ values, in four chains of 5,000. Find the first and second values of λ produced by the first chain. Calculate the likelihood of both of those values, given the data. Use these likelihoods to describe how/why the Metropolis Hastings algorithm moved from the first value to the second value.

```
as.array(gp_sim, pars = "lambda") %>%
  head(4)
```

```
, , parameters = lambda
```

```
      chains
iterations chain:1 chain:2 chain:3 chain:4
[1,] 1.6286360 2.147167 1.899411 1.655500
[2,] 0.9661033 1.127206 1.892485 1.996214
[3,] 1.3867649 1.083150 1.253871 1.859178
[4,] 1.7779811 1.551773 1.344669 1.472758
```

```
prod(dpois(c(1,3,0,4,0,2,2,0,3,1),1.6286))
```

```
[1] 5.991523e-08
```

```
prod(dpois(c(1,3,0,4,0,2,2,0,3,1),.9661))
```

```
[1] 1.061844e-08
```

The first value is 1.6286, and the second value is .9661. The likelihood is higher for the first value than the second value. Therefore the algorithm wouldn't have automatically moved from the first to the second. Rather, an acceptance probability less than one would have been calculated, and based on the result of random weighted coin flip with that probability, it was decided to move to the second value.